

Algorithms and Convergence

① Rate of Convergence for Sequences :

Formula : $|\alpha_n - \alpha| \leq K|\beta_n|$, for large n , $\beta_n = \frac{1}{n^p}$,

6. Find the rates of convergence of the following sequences as $n \rightarrow \infty$.

a. $\lim_{n \rightarrow \infty} \sin \frac{1}{n} = 0$

to solve Q use $\Rightarrow$

$$|\sin x| \leq |x|, \text{ where } 0 \leq x < 1$$

$$= \left| \sin \frac{1}{n} \right| \leq \left| \frac{1}{n} \right|$$

$k=1 \rightarrow \beta_n = \frac{1}{n}$

* by using formula *

$$\therefore \lim_{n \rightarrow \infty} \sin \frac{1}{n} = O\left(\frac{1}{n}\right)$$

b. $\lim_{n \rightarrow \infty} \sin \frac{1}{n^2} = 0$

$$\left| \sin \frac{1}{n^2} \right| \leq \left| \frac{1}{n^2} \right|$$

$k=1 \rightarrow \beta_n = \frac{1}{n^2}$

$$\therefore \lim_{n \rightarrow \infty} \sin \frac{1}{n^2} = O\left(\frac{1}{n^2}\right)$$

c. $\lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right)^2 = 0$

$$= \left| \left(\sin \frac{1}{n} \right)^2 \right| \leq \underbrace{\left| \left(\frac{1}{n} \right)^2 \right|}_{k=1} = \underbrace{\left(\frac{1}{n^2} \right)}_{\hookrightarrow B_n}$$

note: Squar and Absolut value
Can Cancel each other

$$\text{so } \lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right)^2 = O\left(\frac{1}{n^2}\right)$$

d. $\lim_{n \rightarrow \infty} [\ln(n+1) - \ln(n)] = 0$

to solve Q use $\Rightarrow$

$$\left| \ln(1+x) \right| \leq |x|, \text{ where } 0 < x < 1$$

$$= [\ln(n+1) - \ln(n)] = \ln \frac{n+1}{n} = \ln \left(\frac{n}{n} + \frac{1}{n} \right) = \ln \left(1 + \frac{1}{n} \right)$$

$$\Rightarrow \left| \ln \left(1 + \frac{1}{n} \right) \right| \leq \underbrace{\left| \left(\frac{1}{n} \right) \right|}_{k=1 \hookrightarrow B_n}$$

$$\ln(ab) = \ln(a) + \ln(b)$$

$$\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$$

$$\ln(a^b) = b \ln(a)$$

$$\text{so } \lim_{n \rightarrow \infty} [\ln(n+1) - \ln(n)] = O\left(\frac{1}{n}\right)$$

② Rate of Convergence for function

Formula: $|F(h) - L| \leq K|G(h)| \Rightarrow F(h) = L + O(G(h)), \text{ where } G(h) = h^p,$

Maclurin Series

$$\sin h = h - \frac{h^3}{3!} + \frac{h^5}{5!} + \dots$$

$$\frac{n}{h}$$

need the first two values

$$\frac{\cosh h}{h} = \frac{1 - \frac{h^2}{2} + \frac{h^3}{3} + \dots}{h}$$

$$\frac{e^h}{h} = \frac{1 + \frac{h^2}{2} + \frac{h^3}{3} + \dots}{h}$$

7. Find the rates of convergence of the following functions as $h \rightarrow 0$.

a. $\lim_{h \rightarrow 0} \frac{\sinh h}{h} = 1 \rightarrow L$

$$= \frac{\sinh h}{h} = \frac{h - \frac{h^3}{6} + \frac{h^5}{120}}{h} = \frac{h}{h} - \frac{h^2}{6} = 1 - \frac{h^2}{6} \quad \text{use the first (h)}$$

$$\text{so } \lim_{h \rightarrow 0} \frac{\sinh h}{h} = L + O(h^p) = 1 + O(h^2)$$

b. $\lim_{h \rightarrow 0} \frac{1 - \cosh h}{h} = 0 \rightarrow L$

$$\frac{\cosh h}{h} = \frac{1 - \frac{h^2}{2} + \frac{h^3}{3} + \dots}{h}$$

$$\frac{1 - \cosh h}{h} = \frac{1 - (1 - \frac{h^2}{2} + \frac{h^3}{3} + \dots)}{h} = \frac{\cancel{1} - 1 + \frac{h^2}{2} - \frac{h^3}{3} + \dots}{h} = \frac{\frac{h^2}{2} - \frac{h^3}{3}}{h}$$

$$\frac{1 - \cosh h}{h} = \frac{h}{2} - \frac{h^2}{3}$$

use the first (h)

$$\circ_0 \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0 + O(h) = O(h)$$

c. $\lim_{h \rightarrow 0} \frac{\sin h - h \cos h}{h} = 0 \rightarrow L$

$$\frac{\sin h - h \cos h}{h} = \frac{(h - \frac{h^3}{6} + \dots) - h(1 - \frac{h^2}{2} + \dots)}{h}$$

$$= \frac{(\cancel{h} - \frac{h^3}{6}) (\cancel{-h} + \frac{h^3}{2})}{h} = \frac{-h^3}{6} + \frac{h^3 \times 3}{2 \times 3} = \frac{-h^3 + 3h^3}{6} = \frac{2h^3}{6}$$

take (h^3)

$$\circ_0 \lim_{h \rightarrow 0} \frac{\sin h - h \cos h}{h} = 0 + O(h^3) = O(h^3)$$

d. $\lim_{h \rightarrow 0} \frac{1 - e^h}{h} = -1 \rightarrow L$

$$\frac{e^h}{h} = \frac{1 + \frac{h^2}{2} + \frac{h^3}{3} + \dots}{h}$$

$$= \frac{1 - e^h}{h} = \frac{1 - (1 + \frac{h^2}{2})}{h} = \frac{\cancel{1} - 1 - \frac{h^2}{2}}{h} = \frac{-\frac{h^2}{2}}{h} = \frac{-h}{2}$$

take (h)
w/o sign

$$\circ_0 \lim_{h \rightarrow 0} \frac{1 - e^h}{h} = -1 + O(h)$$